

Isaac Henri Joël Houngue

Software and Research Engineer

📍 Cape Town, South Africa ✉ hjisaac.h@gmail.com 📞 +27 79 548 2528 [in](#) [hjisaac](#) [gh](#) [hjisaac](#)

Summary

Motivated and hardworking Software and Research Engineer with MSc backgrounds in Artificial Intelligence and Computer Science, combining 3+ years of software engineering experience with research engineering experience in machine learning and scientific AI systems. Strong debugging and problem-solving skills; passionate about developing intelligent systems and translating research into usable, simple, effective, robust, and innovative solutions. Proactive and initiative-driven, accustomed to quality-focused environments and ready to take on challenging tasks with confidence and commitment.

Work Experience

Apprentice Research Engineer, InstaDeep – South Africa, hybrid Jul 2025 – Now

- Evaluated fine-tuning strategies for adapting InstaNovo, a transformer-based de novo peptide sequencing model, to glycopeptide sequencing tasks.
- Explored contrastive learning, multi-task learning, knowledge distillation, curriculum learning, and transformer-variant architectures to improve sequencing performance.
- Designed modular, pluggable components inspired by research papers to accelerate hypothesis testing and integration of new modeling approaches.
- Conducted large-scale experiments and ablation studies on architectural choices, training strategies, and model components.
- Designed synthetic glycopeptide datasets and multi-stage training strategies to improve representation learning and transfer to experimental tasks.
- Performed glycopeptide spectral dataset analysis and feature selection studies to understand data characteristics and model limitations.
- Documented research findings through technical reports, visualizations, and presentations.

Backend Software Engineer, Fasfox – Paris, remote Mar 2023 – Sep 2024

- Enhanced the backend API of Concrete Dispatch SaaS, a logistics platform for managing concrete deliveries (Django/DRF).
- Improved a fiber ordering and tracking product for AXIONE (Django/DRF).
- Developed a fault-tolerant daemon for parsing large CSV and eligibility data files for Fiber Dispatch (Django/Ninja).
- Built TMForum compliant endpoints for fiber order handling following interop protocols (Django/Ninja).
- Applied web scraping and AI-powered OCR to extract data from PDFs for real-time validation (Django/DRF).
- Built and maintained a daemon that listens to email inboxes and parses attachments to create objects across databases (Django/DRF).

Frontend Software Engineer, Fasfox – Paris, remote Oct 2021 – Feb 2023

- Maintained and enhanced Concrete Dispatch SaaS PWAs: photo uploads, digital signatures, QR scanning, and interactive graphs (Vue.js/Nuxt.js).
- Developed an advanced slider for concrete volume tracking in the desktop web app (React.js/Next.js).
- Built an efficient concrete calculator for the Concrete Dispatch blog site (Hugo/JavaScript).

Software Engineering Intern, National Civil Aviation Agency – Benin, onsite Aug 2019 – Sep 2019

- Developed a Python/Tkinter inventory management desktop application for tracking electronic equipment.
- Assisted in maintenance of office and IT equipment.

Personal Projects

Run Crucible 	Feb 2026 – Present
• Modular framework for managing and comparing machine learning experiments across training configurations and model architectures. • Tools: Python, PyTorch, Hydra, Docker	
Recommender System 	Oct 2024 – Feb 2025
• Developed a recommender system using ALS and collaborative filtering built using the MovieLens dataset. • Tools: Python, NumPy, Streamlit	
NSequence 	Jan 2024 – Feb 2024
• Developed a Python library for manipulating progressions or sequences. • Tools: Python	
BVote 	Jun 2021 – Jul 2021
• Developed a secure voting platform with anonymity via blind signatures and blockchain for transparency. • Tools: Node.js, ReactJS, GraphQL, HyperLedger, MongoDB	
Stream Processing Engine 	Feb 2021 – Mar 2021
• Built a simple e-commerce stream processing engine. • Tools: Python, Kafka, Redis, Socket.io	
Billvoicer [Private]	Jan 2023 – Now
• Initiator and principal engineer of Billvoicer, driving the project's vision and technical development. • Leading architecture design, feature development, and technology decisions. • Managing a team of interns, overseeing development activities, issue tracking, code reviews and more. • Ensuring seamless team communication and alignment with the product roadmap. • Tools: Python (Django/Drf), VueJS/NuxtJS, PostgreSQL, Redis, Celery	

Technologies

Programming Languages: Python, JavaScript, TypeScript

Frameworks And Libraries: Django, DRF, Flask, Ninja, Node.js, Vue.js, React.js, Next.js, GraphQL, PyTorch, Keras

Infrastructure And Tools: Linux, Git, Docker, GCP, AWS, Kafka, Spark, PostgreSQL, MongoDB, Neo4j

Education

Master in Artificial Intelligence , African Institute for Mathematical Sciences (AIMS) – South Africa	Sep 2024 – Jul 2025
• Coursework: Bayesian Inference, Machine Learning at Scale, Computer Vision, Reinforcement Learning, Deep Generative Models, Engineering LLMs, CUDA Programming, and more. • Thesis: Advancing De Novo Glycopeptide Sequencing with InstaNovo in Glycoproteomics.	
Master in Computer Science , Institute of Mathematics and Physical Sciences (IMSP) – Benin	Oct 2018 – Jul 2020
• Coursework: Advanced Data Structures and Algorithms, Operating Systems, Software Architecture, Distributed Databases, Artificial Intelligence, Machine Learning and more. • Thesis: A Smart Contract-based Remote Voting Platform using Blind Signature.	

Languages

English: Professional proficiency

French: Native

Referees

Lélio Renard Lavaud 

Head of Engineering at Mistral AI, CEO and Chairman at Fasfox

Dr. Anicet Hounkanrin 

PhD in Electrical Engineering, University of Cape Town